

Course 1

Relations



Prof. dr. Septimiu Crivei

Algebra - First year - CS & AI

- Chapter 1: Preliminaries
- Chapter 2: Vector Spaces
- Chapter 3: Matrices and Linear Systems
- Chapter 4: Introduction to Coding Theory

Coordinates: bibliography

-  S. Crivei, *Basic Linear Algebra*, Presa Universitară Clujeană, Cluj-Napoca, 2022.
-  W. J. Gilbert, W. K. Nicholson, *Modern Algebra with Applications*, John Wiley, 2004.
-  J. S. Golan, *The Linear Algebra a Beginning Graduate Student Ought to Know*, Springer, Dordrecht, 2007.
-  P. N. Klein, *Coding the Matrix. Linear Algebra through Applications to Computer Science*, Newtonian Press, 2013.
-  R. Lidl, G. Pilz, *Applied Abstract Algebra*, Springer-Verlag, 1998.
-  I. Purdea, C. Pelea, *Probleme de algebră*, Eikon, Cluj-Napoca, 2008.
-  L. Robbiano, *Linear Algebra for Everyone*, Springer, Milan, 2011.
-  G. Strang, *Linear Algebra and its Applications*, Brooks/Cole, 1988.

- Course materials will be available in MS Teams *Algebra-CS-AI (2025-2026)* (code: 58lf1dy).
- Minimum attendance: 75% for seminar classes in order to be allowed to participate in the second partial exam and pass the course.
- Problems for the next week will be available in MS Teams *Algebra-CS-AI (2025-2026)* (code: 58lf1dy).
- Students may get up to 0.5 bonus points from seminar to the final grade: 5 problems solved during the seminar, each for 0.1 points [details during seminars].

- Written partial exams in:
Week 8 (Chapters 1-2): Saturday, November 22, 2025
Week 14 (Chapters 3-4): Saturday, January 17, 2026
- The final grade is computed as follows:

$$G = 1 + P_1 + P_2 + B,$$

where:

G = the final grade

P_1 = the points from the first partial exam (max. 4.5)

P_2 = the points from the second partial exam (max. 4.5)

B = bonus points from seminar (max. 0.5)

- Students may not pass the exam unless they participate in the second partial exam.

The Association for Computing Machinery (ACM) has developed the 2012 ACM Computing Classification System for the research topics in the field of Computer Science (www.acm.org) under the form of a multi-level tree.

We mention some higher level branches of this tree in which Linear Algebra has important applications.

● **Networks**

- Network architectures
 - Network design principles
- Network types
 - Public Internet

● **Theory of Computation**

- Models of computation
 - Quantum computation theory
- Computational complexity and cryptography
 - Cryptographic protocols

- Randomness, geometry and discrete structures
 - Error-correcting codes
- Theory and algorithms for application domains
 - Machine learning theory

● **Mathematics of Computing**

- Information theory
 - Coding theory
- Mathematical analysis
 - Mathematical optimization

● **Information Systems**

- World Wide Web
 - Web searching and information discovery
- Information retrieval
 - Retrieval models and ranking

● **Security and Privacy**

- Cryptography
 - Symmetric cryptography and hash functions

- Network security
 - Security protocols

● **Computing Methodologies**

- Machine learning
 - Machine learning approaches
- Computer graphics
 - Image manipulation

● **Applied Computing**

- Electronic commerce
 - Online banking
- Operations research
 - Decision analysis

Chapter 1. Preliminaries

- 1 Relations
- 2 Functions
- 3 Equivalence relations and partitions

Application: relational database

ID (Integer)	Surname (String)	Name (String)	Grade (Integer)
7	Ionescu	Alina	9
11	Ardelean	Cristina	10
23	Ionescu	Dan	7

Definition

A triple $r = (A, B, R)$, where A, B are sets and

$$R \subseteq A \times B = \{(a, b) \mid a \in A, b \in B\},$$

is called a *(binary) relation*.

The set A is called the *domain*, the set B is called the *codomain* and the set R is called the *graph* of the relation r .

If $A = B$, then the relation r is called *homogeneous*.

If $(a, b) \in R$, then we sometimes write $a r b$ and we say that a *has the relation r to b* or a and b *are related with respect to the relation r* .

Definition

Let $r = (A, B, R)$ be a relation and let $X \subseteq A$. Then the set

$$r(X) = \{b \in B \mid \exists x \in X : x r b\}$$

is called the *relation class of X with respect to r* .

If $x \in X$, then we denote

$$r \langle x \rangle = r(\{x\}) = \{b \in B \mid x r b\}.$$

Notice that

$$r(X) = \bigcup_{x \in X} r \langle x \rangle .$$

Relation representation

If A, B are finite sets, then $r = (A, B, R)$ may be represented by a diagram consisting of two sets with elements and connecting arrows. For instance, let $r = (A, B, R)$, where $A = \{1, 2, 3\}$, $B = \{1, 2\}$ and

$$R = \{(1, 1), (1, 2), (3, 1)\}.$$

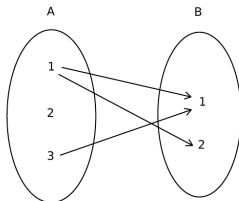


Figure: Diagram of a relation.

Also note that $r \circ 1 \supseteq \{1, 2\} = r(A)$.

Examples of relations I

(a) Let C be the set of all children and let P be the set of all parents. Then we may define the relation $r = (C, P, R)$, where

$$R = \{(c, p) \in C \times P \mid c \text{ is a child of } p\}.$$

(b) The triple $r = (\mathbb{R}, \mathbb{R}, R)$, where

$$R = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x \leq y\}$$

is a homogeneous relation, called the *inequality relation* on $\mathbb{R}$. We have

$$r < 1 \supseteq [1, \infty) = r([1, 2]).$$

(c) There are several examples from Number Theory, such as divisibility on $\mathbb{N}$ or on $\mathbb{Z}$, and Geometry, such as parallelism of lines, perpendicularity of lines, congruence of triangles, similarity of triangles.

Examples of relations II

(d) Let A and B be two sets. Then the triples

$$o = (A, B, \emptyset), \quad u = (A, B, A \times B)$$

are relations, called the *void relation* and the *universal relation* respectively.

(e) Let A be a set. Then the triple $\delta_A = (A, A, \Delta_A)$, where

$$\Delta_A = \{(a, a) \mid a \in A\}$$

is a relation called the *equality relation* on A .

(f) Every function is a relation. Indeed, a function $f : A \rightarrow B$ is determined by its domain A , its codomain B and its graph

$$G_f = \{(x, y) \in A \times B \mid y = f(x)\}.$$

Then the triple (A, B, G_f) is a relation.

Examples of relations III

(g) Every directed graph is a relation.
For instance, the directed graph

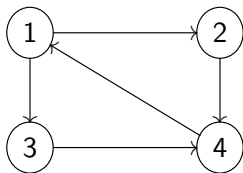


Figure: Directed graph.

may be seen as a relation (A, A, R) , where $A = \{1, 2, 3, 4\}$ and

$$R = \{(1, 2), (1, 3), (2, 4), (3, 4), (4, 1)\}.$$

Definition

A relation $r = (A, B, R)$ is called a *function* if

$$\forall a \in A, \quad |r \langle a \rangle| = 1,$$

that is, the relation class with respect to r of every $a \in A$ consists of exactly one element.

In what follows, if $f = (A, B, F)$ is a function, we will mainly use the classical notation for a function, namely $f : A \rightarrow B$ or sometimes $A \xrightarrow{f} B$. The unique element of the set $f \langle a \rangle$ will be denoted by $f(a)$. Then we have

$$(a, b) \in F \iff f(a) = b.$$

Functions - related notions

From relations we get the following notions.

Definition

Let $f : A \rightarrow B$ be a function. Then A is called the *domain*, B is called the *codomain* and

$$F = \{(a, f(a)) \mid a \in A\}$$

is called the *graph* of the function f .

Definition

Let $f : A \rightarrow B$ be a function and let $X \subseteq A$. We call the *image of X by f* the relation class of X with respect to f , that is,

$$f(X) = \{b \in B \mid \exists x \in X : x f b\} = \{f(x) \mid x \in X\}.$$

We denote $\text{Im}f = f(A)$ and call it the *image of f* .

Examples of functions and relations

(a) Let A be a set. Then the equality relation (A, A, Δ_A) is a function called the *identity function on A* , denoted by $1_A : A \rightarrow A$.

(b) Let $A = \{1, 2, 3\}$, $B = \{1, 2\}$ and let $r = (A, B, R)$, $s = (A, B, S)$, $t = (A, B, T)$ be the relations having the graphs

$$R = \{(1, 1), (2, 1), (3, 2)\},$$

$$S = \{(1, 2), (3, 1)\},$$

$$T = \{(1, 1), (1, 2), (2, 1), (3, 2)\}.$$

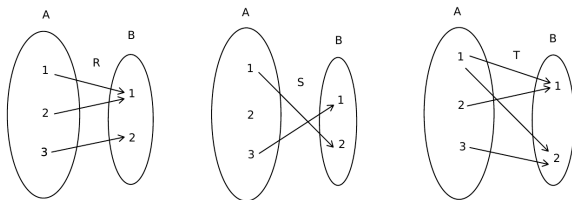


Figure: Diagrams of functions or relations.

Which of them are functions?

Recall that a relation $r = (A, B, R)$ is called *homogeneous* if $A = B$.

Definition

A homogeneous relation $r = (A, A, R)$ on A is called:

- (1) *reflexive* (r) if: $\forall x \in A, x r x$.
- (2) *transitive* (t) if: $x, y, z \in A, x r y$ and $y r z \implies x r z$.
- (3) *symmetric* (s) if: $x, y \in A, x r y \implies y r x$.

A homogeneous relation $r = (A, A, R)$ is called an *equivalence relation* if r has the properties (r), (t) and (s).

Examples of equivalence relations

- (a) The equality relation δ_A on a set A is an equivalence relation.
- (b) The similarity of triangles is an equivalence relation on the set of all triangles.
- (c) The inequality relation “ $\leq$ ” on $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$ has (r) and (t), but not (s). Hence it is not an equivalence relation.
- (d) Let $n \in \mathbb{N}$ and let ρ_n be the relation defined on $\mathbb{Z}$ by

$$x \rho_n y \iff x \equiv y \pmod{n},$$

that is, $n|(x - y)$ or equivalently for $n \neq 0$, x and y give the same remainder when divided by n . Then ρ_n is called the *congruence modulo n* and it is an equivalence relation.

For $n = 0$, we have $x \rho_0 y \iff 0|x - y \iff x = y$, hence $\rho_0 = \delta_{\mathbb{Z}} = (\mathbb{Z}, \mathbb{Z}, \Delta_{\mathbb{Z}})$.

For $n = 1$, we have $x \rho_1 y \iff 1|x - y$, which is always true, and thus $\rho_1 = u = (\mathbb{Z}, \mathbb{Z}, \mathbb{Z} \times \mathbb{Z})$.

Definition

Let A be a non-empty set. Then a family $(A_i)_{i \in I}$ of non-empty subsets of A is called a *partition* of A if:

(i) The family $(A_i)_{i \in I}$ covers A , that is,

$$\bigcup_{i \in I} A_i = A.$$

(ii) The A_i 's are pairwise disjoint, that is,

$$i, j \in I, i \neq j \implies A_i \cap A_j = \emptyset.$$

Examples of partitions

(a) Let $A = \{1, 2, 3, 4, 5\}$ and $A_1 = \{1, 2, 3\}$, $A_2 = \{4\}$, $A_3 = \{5\}$. Then $\{A_1, A_2, A_3\}$ is a partition of A .

(b) Let A be a set. Then $\{\{a\} \mid a \in A\}$ and $\{A\}$ are partitions of A .

(c) Let A_1 be the set of even integers and A_2 the set of odd integers. Then $\{A_1, A_2\}$ is a partition of $\mathbb{Z}$.

(d) Consider the intervals

$$A_n = [n, n + 1)$$

for every $n \in \mathbb{Z}$. Then the family $(A_n)_{n \in \mathbb{Z}}$ is a partition of $\mathbb{R}$.

Denote by $E(A)$ the set of all equivalence relations and by $P(A)$ the set of all partitions on a set A .

Definition

Let $r \in E(A)$.

The relation class $r < x >$ of an element $x \in A$ with respect to r is called the *equivalence class of x with respect to r* , while the element x is called a *representative* of $r < x >$.

The set

$$A/r = \{r < x > \mid x \in A\},$$

which is the set of all equivalence classes of elements of A with respect to r , is called the *quotient set of A by r* .

Relation associated to a partition

Definition

Let $\pi = (A_i)_{i \in I} \in P(A)$ and define the relation r_π on A by

$$x r_\pi y \iff \exists i \in I : x, y \in A_i.$$

Then r_π is called the *relation associated to the partition* π .

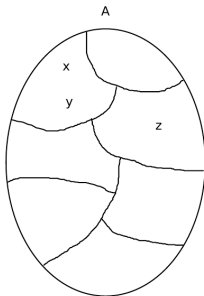


Figure: Relation associated to a partition.

Theorem

- (i) Let $r \in E(A)$. Then $A/r \in P(A)$.
- (ii) Let $\pi = (A_i)_{i \in I} \in P(A)$. Then $r_\pi \in E(A)$.
- (iii) Let $F : E(A) \rightarrow P(A)$ be defined by

$$F(r) = A/r, \quad \forall r \in E(A).$$

Then F is a bijection, whose inverse is $G : P(A) \rightarrow E(A)$, defined by

$$G(\pi) = r_\pi, \quad \forall \pi \in P(A).$$

Illustrations of the theorem I

(a) Let $A = \{1, 2, 3\}$ and let r and s be the homogeneous relations defined on A with the graphs

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\},$$

$$S = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}.$$

Then r is an equivalence relation, but s is not. What is the partition corresponding to r ?

(b) Consider the following families of sets:

$$\pi = \{\{1\}, \{2, 3\}, \{4\}\},$$

$$\pi' = \{\{1, 2\}, \{2, 3\}, \{4\}\}.$$

Then π is a partition of $A = \{1, 2, 3, 4\}$, but π' is not. What is the equivalence relation corresponding to π ?

Illustrations of the theorem II

(c) The congruence relation modulo n is an equivalence relation on $\mathbb{Z}$ and its corresponding partition is

$$\mathbb{Z}/\rho_n = \{\rho_n < x > \mid x \in \mathbb{Z}\} = \{x + n\mathbb{Z} \mid x \in \mathbb{Z}\} = \{\hat{x} \mid x \in \mathbb{Z}\},$$

where an equivalence class is denoted by $\hat{x}$. For $n \geq 2$, we denote

$$\mathbb{Z}_n = \mathbb{Z}/\rho_n = \{\hat{0}, \hat{1}, \dots, \widehat{n-1}\}.$$

For $n = 0$ and $n = 1$, we have seen that $\rho_0 = \delta_{\mathbb{Z}}$ and $\rho_1 = u$, and we get

$$\mathbb{Z}/\rho_0 = \{\{x\} \mid x \in \mathbb{Z}\} \quad \text{and} \quad \mathbb{Z}/\rho_1 = \{\mathbb{Z}\},$$

that are the two extreme partitions of $\mathbb{Z}$.

Binary relations may be naturally generalized as follows.

Definition

A (finite) tuple

$$r = (A_1, \dots, A_n, R),$$

where $A_1, \dots, A_n$ are sets and

$$R \subseteq A_1 \times \dots \times A_n = \{(a_1, \dots, a_n) \mid a_1 \in A_1, \dots, a_n \in A_n\},$$

is called an *(n-ary) relation*.

The sets $A_1, \dots, A_n$ are called the *domains* of r , and the set R is called the *graph* of r .

The number n is called the *degree (arity)* of r .

A *relational database* is a (finite) set of relations.

Extra: Relational database II

Consider the relation

$$student = (Integer, String, String, Integer, Student),$$

where

$$Student \subseteq Integer \times String \times String \times Integer$$

is given by the following table:

ID (Integer)	Surname (String)	Name (String)	Grade (Integer)
7	Ionescu	Alina	9
11	Ardelean	Cristina	10
23	Ionescu	Dan	7

Some known relational database management systems are:

- Oracle and RDB - Oracle
- SQL Server and Access - Microsoft